Engineering & Applied Science
UNIVERSITY OF COLORADO **BOULDER**

T3am Thr33: Smart-Tongs

A Food-Waste Reduction Product

Created In Team by Max G, Nikky W, Jake S, David G, Declan O.

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How Might We Reduce Food Waste?

***Define**: Students Need a way to be mindful of their consumption habits.*

- *It's too easy to take more food than you can eat*
- *Leftovers end up getting wasted*
- *Current consumption habits are not sustainable*

***Ideate**: How can we effectively influence consumption habits.*

Qualitative:

- Ease of use
- Easy to understand feedback
- Simple to understand how to use

Quantitative:

- Able to handle a grip of 125 lbs of force)
- Spring constant k = 1 N/mm to not be too strong or weak

Functional:

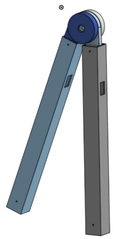
- Able to open and close well.
- Able to pick up all possible food with relative ease.
- Able to provide feedback.

Bill of Materials

Name:	Material:	Price:
Hinge	ABS plastic	ABS price
User feedback	Mixed	$16.00
Grippers	ABS plastic	ABS price
Rumble Motors	Mixed	$7.52
PCB	PCB	$96.00
SSD 1306 Display	Mixed	$7.36
Box Extrusion	Aluminum 6061	$21.88
Accelerometer mpu6050	Mixed	$10.99
Custom Circuit Boards	Fiberglass	$91.58
ABS Filiment	ABS plastic	$23.00

***The Concept:** Smart-Tongs that provide consumption feedback.*

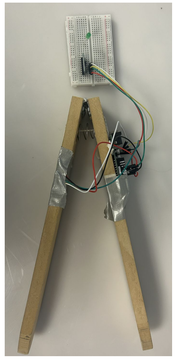
- **Record Serving Size**
  - Live-Axle / Sensor Measured Hinge
  - Inertial Measurement Unit (Movement Sensor)
  - 240 Mhz Dual Core Processor
- **Report Excessive Consumption**
  - Micro OLED-Display
  - Rumble Motor
  - Piezo Element Buzzer
- **Relay Consumption Statistics to Base**
  - Wireless Connectivity over WIFI
  - 2.4GHz Short Range Antenna
  - Bluetooth-LE Support



Prototype Progression

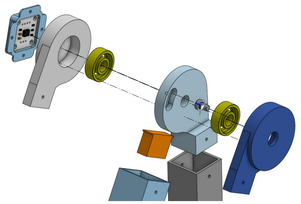
***REV 0-1**: Pretotypes*

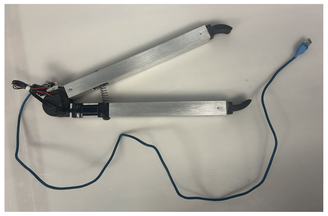
- *Quick and Simple*

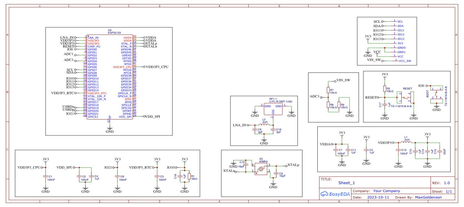


***REV 2**: Feature Complete Prototype*

- *Measures Consumption*
- *Reports Feedback*

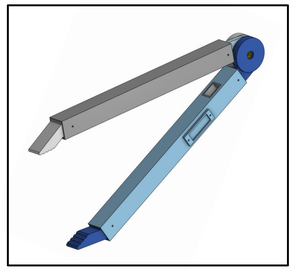


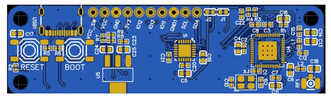




***REV 3**: High Fidelity Prototype*

- *Fully Functional*
- *Wireless Connectivity*
- *Internal Battery*
- *Custom Circuit Board*





Analysis: Tongs and Team

Team Growth

- Each team member has grown their skills in specific areas such as coding, CAD, or machining as the project has progressed.
- Understanding teammates' strengths and weaknesses.
- Communication skills have improved, teamwork with other engineers has become more natural.
- Understanding of the design process of projects.

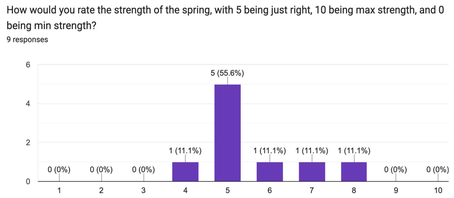
Future Work

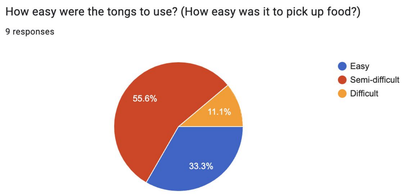
Digitalizing the way we serve ourselves food, while making people aware of the consequences of our actions to our environment.

To take the product further, ideas of other kitchen utensils have been considered, such as ladles, serving spoons, or plates.

Testing Results:

To test the product, we are using multiple tongs, one set with user feedback and one without, in order to determine whether or not the user feedback aided in the mitigation of food waste.





Acknowledgements

This project has been exceptional in bringing out the creative side of all of us, while binding us together as a team and most importantly, as friends. We feel that this class was very efficient in teaching us the engineering skills that are fundamental for our careers in engineering. We are very grateful for all the help and support from our professors and ITLL staffs, without them this project would not have been a success.

Materials/Budget

Part Name:	Quantity:	Material:	Bought/Manufactured?	PPU:
Tong Sides	2	Aluminum 6061	Manufactured	$1.61
Back Hinge	1	PLA	Manufactured	$0.00
Middle Hinge	1	PLA	Manufactured	$0.00
Front Hinge	1	PLA	Manufactured	$0.00
Electronics Holder	1	PLA	Manufactured	$0.00
Bearing Insert	1	PLA	Manufactured	$0.00
Hinge Lock Bolt	1	Steel	Bought	(Consumable)
Rotary Encoder	1	Mixed	Bought	(Consumable)
Dipole Magnet	1	Mixed	Bought	(Consumable)
Hinge Screws	4	Steel	Bought	(Consumable)
Battery	1	Mixed	Bought	$3.99
Accelerometer	1	Mixed	Bought	$3.33
Tong ends	2	PLA	Manufactured	$0.00
Spring mounts	2	PLA	Manufactured	$0.00

Design Requirements:

Qualitative:	Quantitative:	Functional:
• Ease of use • Easy to understand feedback • Simple to understand how to use	• Able to handle a grip of 125 lbs of force) • Spring constant k = 1 N/mm to not be too strong or weak	• Able to open and close well. • Able to pick up all possible food with relative ease. • Able to provide feedback.

Design Constraints

Name	Dimensions (L, W, H) (mm)	Max-min temps (C)	Power Draw	Food/clean Safe
Hinge	(<60, <60, <60)	0-60	no	yes
Hinge shaft	(>40, 8, 8)	0-60	no	yes
Magnet	(2, 4, 4)	0-60	no	no
Encoder	(23, 23, 3.5)	0-60	yes ( )	no
Grippers	(<20, <20, <30)	0-60	no	yes
Tong shafts	(200<x<300, 20<x<30, 20<x<30)	0-60	no	yes
PCB	(75, 20, 3.5)	0-60	yes ( )	no
Accelerometer	(<23, <23, <23)	0-60	yes ( )	no
Spring	(<12, <12, 90<x<100) .5<k<1.5	0-60	no	yes
Bearings	(7, 22, 22)	0-60	no	yes
Rumble Motors	11mm x 3mm (D x T)	0-60	yes ( )	no
Screws	M2 x 4, M4x4	0-60	no	yes
Batteries	(18, 18, 66)	0-61	yes ( )	no

Smart Tongs Concept

- **Record Serving Size**
  - Live-Axle / Sensor Measured Hinge
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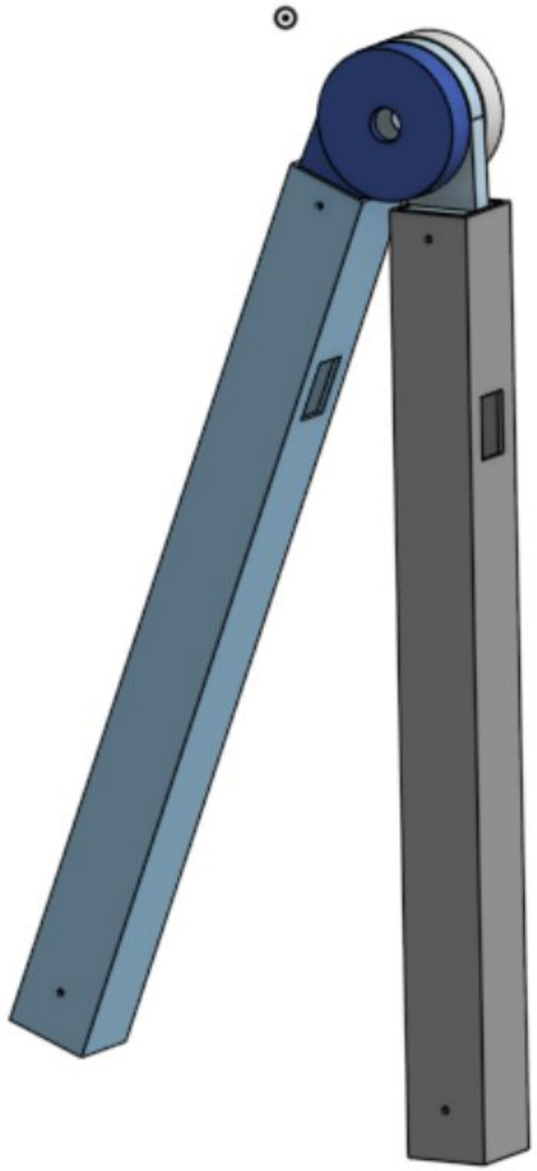
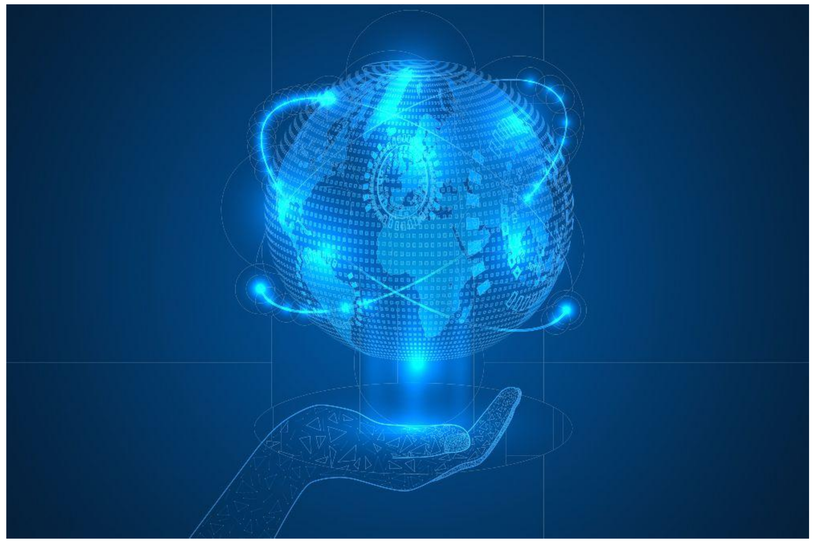


Future Work



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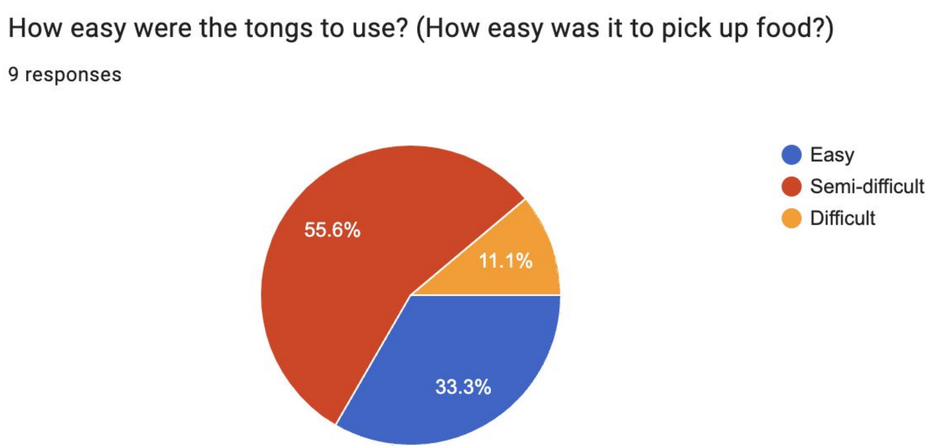

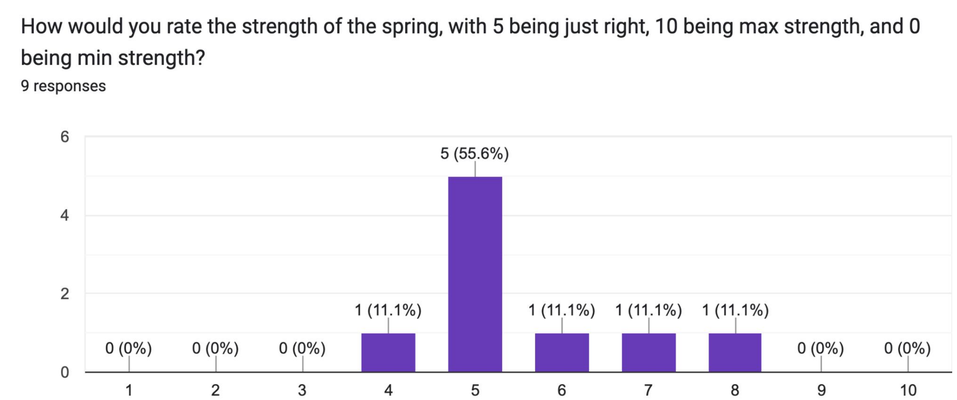

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